FLEET TEAM DIGEST

737MAX-FTD-21-21004

Issue Title : Cabin Altitude Pressure Switch Dual Failure

Airplane Model	ATA	Minor Model(s)
737MAX	2133-04	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG, 737	9E991

Originated Date	Last Revised Date	Created On
08/02/2021	03/13/2024	09/27/2023

Estimated Completion Date	Next Update Date
08/29/2025	06/19/2024

Revision Description

Updated the Status, Final Actions, and Milestones sections within this FTD article to reflect the latest project status.

All changes are highlighted in blue text.

Description

In August 2020, Boeing received a report of dual failures of the cabin altitude pressure switches which occurred during performance of a required Maintenance Planning Document (MPD) test procedure. The inability to pass the functional test represents a latent failure of the cabin altitude pressure switches. A failure of both pressure switches could result in the cabin altitude warning system not activating if the cabin altitude exceeded 10,000 ft.

Background

The 737 MAX utilizes the same cabin altitude warning switches as the 737NG in the similar cabin altitude warning system design. The cabin altitude warning system is comprised of two identical aneroid pressure switches, the P5-16 flight deck panel that provides a cutout button for the intermittent horn, an aural warning module that creates the intermittent horn sound, and two CABIN ALTITUDE red warning level indicators and interconnecting wiring. The Cabin Altitude Warning system annunciates at 10,000 feet nominally with the red CABIN ALTITUDE indicators and the intermittent horn. When the pressure switch fails, it fails latently; that is, without making the failure known to the flight crew or maintenance personnel.

The MPD requires a functional test of the cabin altitude pressure switches to be performed at 6,000 flight hour (FH) intervals. When a pressure switch is replaced, the Airplane Maintenance Manual (AMM) installation task requires a functional check to be performed on both switches. This ensures that spare (serviceable) pressure switches are functioning properly when installed.

Following the operator report in August 2020, a Continued Operational Safety Program (COSP) investigation was opened (COSP 2020-1180). The initial investigation did not lead to a finding of Airplane Safety based on data and analysis available at the time. However, a subsequent investigation and analysis led the FAA and Boeing to determine, in May 2021, that the failure rate of both switches was much higher than initially estimated, and therefore did pose a safety issue.
One finding during the investigation was that removal data available through Boeing's In Service Data Program (ISDP) did not

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indicate a high rate of failure of the switches. Further review also determined that the pressure switch has been treated as a consumable by many operators, and the removed switches were not returned to the supplier for any troubleshooting. Therefore, the actual removal rates were found to be higher than expected.

Pressure switches removed in 2020 during Task Card checks were returned to the supplier, Eaton Aerospace, for testing and tear-down. Units were found to pass the incoming acceptance tests as No Fault Found (NFF).

A Safety Service Related Problem (SRP) project, 737NG-SRP-21-0280, was initiated on 20-May-2021 (applicable to both 737NG / 737 MAX) where Boeing coordinated with the FAA to support the release of Airworthiness Directive (AD) 2021–14–20. The FAA AD decreased the pressure switch test interval to 2,000 flight hours and required operators to submit functional test failure reports. Boeing collected in-service data on the failure rate and worked with the supplier, Eaton Aerospace, to determine the root cause of the issue.

Status

Boeing issued Multi-Operator-Message MOM-MOM-21-0292-01B on 23-Jun-2021, to inform operators of the component issue and the recommended interim actions. The FAA issued an Airworthiness Directive (AD) 2021-14-20 on 20-Jul-2021 to require action by operators.

During the root cause investigation, Boeing learned that one operator had experienced test failures that were attributed to unclear instructions on use of the test equipment adapter fittings. The operator found that the test adapter had bottomed-out inside of the pressure switch port and that had prevented the test equipment from achieving a proper vacuum on the pressure switch. This prevented the pressure switch from responding to changes in vacuum pressure, and error indications were not present on the air data tester.

Consequently, as the altitude was increased by the air data tester, the pressure switch did not actuate at any altitude. Boeing determined that using a wrench to tighten the adapter could cause the adapter to seal at the bottom of the pressure port, blocking vacuum pressure to the switch and causing false test failures. Hand tightening the adapter or installing fully and backing out one full turn helped to prevent false test failures. This condition and the potential to interfere with switch operation was verified during testing at Boeing's lab. Boeing issued a revision to the Aircraft Maintenance Manual (AMM 21-33-00-000-801) to improve instructions on use of adapters, as described in Service Letter 737-SL-21-146.

Additionally, Boeing submitted an Alternative Means of Compliance (AMOC) letter to allow operators to send the 30-day interval report up to 5 business days after the 30th day, and allow such report to include only the airplanes that performed the initial functional test within the time period of the 30-day interval report. A Global AMOC Reference number 785-21-13854 was released on 18-Oct-2021 and can be found within MyBoeingFleet.

Boeing tracked pressure switch removals and supplier test results. Eaton completed the teardown and examination of failed units and determined there was no common mode of failure. The Root Cause was therefore confirmed to be that AMM instructions (Task 21-33-00-000-801) on use of test equipment adapter fittings were unclear and could result in false test failures if adapter bottoms out and blocks sensing holes.

Boeing met with the FAA to present the investigation findings and discuss the goal of restoring the 6000 FH inspection interval. A Transport Airplane Risk Assessment Methodology (TARAM) analysis was performed, using the in-service failure data from the pressure switch investigation. When applying the 6000 Flight Hour (FH) inspection interval to the analysis, it was determined that the TARAM calculation exceeded the TARAM guidance value. As a result, the AD requirement to perform the pressure switch functional test at a 2000 FH interval will be maintained.

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The FAA issued a superseding AD 2022-24-09 (same as AD 2021-14-20, but without operators' reporting requirements). The pressure switch functional test interval will be maintained at 2000 FH per the AD requirement. Boeing will perform another review of pressure switch failure data in approximately 18 months to determine if the test interval can be increased to above the current 2000 FH requirement.

Interim Action

None.

Final Action

Boeing will perform another review of pressure switch failure data in approximately 18 months to determine if the test interval can be increased to above the current 2000 FH requirement.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.
737NG-SRP-21-0280 Opened: 5/20/2021 (complete)
737-SRP-21-0280 Opened: 5/20/2021 (complete)
Lead Airline Call: 6/17/2021 (complete)
Fleet Team Call: 6/22/2021 (complete)
MOM-MOM-21-0292-01B Release: 6/23/2021 (complete)
FAA AD 2021-14-20 Release: 7/20/2021 (complete)
Fleet Team Call: 9/16/2021 (complete)
737NG AMM-TR Release: 9/17/2021 (complete)
Service Letter 737-SL-21-146 Release: 9/30/2021 (complete)
AMOC 785-21-13854 Released: 10/18/2021 (complete)
Fleet Team Meeting: 10/26/2021 (complete)
Fleet Team Call: 12/09/2021 (complete)
Data Collection Complete: 01/07/2022 (complete)
Root Cause: 01/07/2022 (complete)
Superseding FAA AD 2022-24-09 Released: 01/26/2023 (complete)
Boeing updated review of failure data: Q3 of 2025 (target)

Operator Action

Operators will continue to perform the cabin altitude pressure switch functional test at the 2000 FH interval per the superseding AD 2022-24-09 requirement. The reporting requirement is no longer in effect.

Boeing also recommends that operators use the improved test equipment set-up procedures to help prevent false test failures.

Part Information

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Boeing initiated a parts and tool availability assessment/planning action to ensure there is an adequate supply of spares in stock (P/N 214C50-2) to support the fleet. Boeing worked with Eaton Aerospace to increase production and have sufficient parts available on time to support pressure switch functional tests.
Boeing requests that operators please work with your Field Service team when requesting parts.

Supplier Information

Eaton Aerospace

Fluid & Electrical Distribution Div

15 Durant Avenue Bethel, CT 06801

Phone: 203-796-6000

Fax: 203-796-6313

Toll Free: 800-736-1557

Attn: Jill Elko

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-21-21002		
Other	AMOC 785-21-13854		10/18/2021
Multi-Operator Message (MOM)	MOM-MOM-21-0292-01B		
Airworthiness Directive (AD)	2021-14-20		
Component Maintenance Manual (CMM)	21-26-01		
Service Related Problem (SRP)	737NG-SRP-21-0280		
Maintenance Planning Document (MPD)	31-020-00		
Drawing	214C50-2 Pressure Switch		
Maintenance Check Manual	TASK CARD 31-020-00-01		
Aircraft Maintenance Manual (AMM)	21-33-00-000-801		
Service Letter (SL)	737-SL-21-146		
Airworthiness Directive (AD)	2022-24-09		

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Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Pressure Switch	10-2638-4	Eaton Aerospace	214C50-2

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- SRPs - Safety : Service Related Problems in work by Boeing that have an airplane or personal safety impact

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